BALANCED RANDOM SUBSET SUMS IN FINITE ABELIAN GROUPS: EXPLICIT CORRECTIONS OF ORDER $\sqrt{q \log q}$

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ABSTRACT. Let G be any finite abelian group of order N, and put $q = \log_2 N$. We give explicit sufficient sample sizes for all subset-sum representation counts to lie within relative error $1/2$ of $2^k/N$. For independent uniform samples, every integer $k \geq q + 8400\sqrt{q \log_2 q}$ has failure probability at most $1/(2q)$ when $N \geq 2^{64}$. The constant is 200 when $N \geq 2^{486018750}$. For a uniform k-element subset the failure bound is $1/q$, provided also $k \leq \sqrt{N}$. We give two intermediate explicit bounds and a corollary with $k = \lceil q + 3q^{4/5} \rceil$ at the larger onset. The proof splits a Fourier power according to integer relation lattices, bounds the low-rank part deterministically on a common event, and takes an unconditional second moment of the remaining polynomial. A count of bounded-generator lattices yields the square-root correction. The leading-coefficient-one question is already known to have an affirmative answer; the subject here is its explicit correction, uniformly over all finite abelian groups.

1. INTRODUCTION AND STATEMENTS

Write the operation of a finite abelian group G additively, and let $N = |G|$. For a tuple $A = (g_1, \dots, g_k)$ define

$$R_A(g) = \#\{S \subseteq \{1, \dots, k\} : \sum_{i \in S} g_i = g\}.$$

The empty subset has sum zero. Repeated entries retain distinct indices. In model (I), the entries are independent and uniform in G. In model (U), A is a uniform k-element subset of G, and R_A counts subsets of this set. In either model put

$$q = \log_2 N, \qquad L = \log_2 q, \qquad D(g) = \frac{N R_A(g)}{2^k} - 1, \qquad \mathcal{F} = \{\max_{g \in G} |D(g)| > 1/2\}.$$

Thus "balanced" means the non-strict inequality $\max_g |D(g)| \leq 1/2$. The numbers q and L are real, and are never assumed integral. Every probability assertion below is for each fixed group separately, with the same bound for every group of the given order.

The question recorded as Erdős Problem 1179 [1] asks for leading coefficient one in the sample size for near-equal subset-sum counts, with high probability. Erdős and Hall [3] already settled that question. We study explicit smaller corrections. We retain the following two intermediate results to display the successive parameter choices; Theorem 3 is the main estimate.

Theorem 1. *Let G be any finite abelian group of order $N \geq 2^{2^{20}}$, and let*

$$k = \left\lceil q + \frac{1000q}{\log_2 q} \right\rceil.$$

Then $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \leq 1/(2q)$. Moreover $k \leq N$, and $\mathbb{P}_{\mathrm{U}}(\mathcal{F}) \leq 1/q$.

Theorem 2. *Let G be any finite abelian group of order $N \geq 2^{64}$, and let $k = \lceil q + 3400q^{4/5} \rceil$. Then $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \leq 1/(2q)$. Moreover $k \leq N$, and $\mathbb{P}_{\mathrm{U}}(\mathcal{F}) \leq 1/q$.*

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Theorem 3. *Let G be any finite abelian group of order N.*

(a) *If $N \ge 2^{64}$, every integer $k \ge q + 8400\sqrt{qL}$ satisfies $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le 1/(2q)$. If also $k \le 2^{q/2}$, then $\mathbb{P}_{\mathrm{U}}(\mathcal{F}) \le 1/q$.*

(b) *If $N \ge 2^{486018750}$, the same assertions hold with 200 in place of 8400, retaining the condition $k \le 2^{q/2}$ for model* (U).

The admissible integer interval in model (U) *is nonempty in both parts.*

Corollary C. *For every finite abelian group of order $N \ge 2^{486018750}$ and $k = \lceil q + 3q^{4/5} \rceil$, one has $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le 1/(2q)$ and $\mathbb{P}_{\mathrm{U}}(\mathcal{F}) \le 1/q$.*

There are no asymptotic error terms in these statements or their proofs. For the minimal rounded size in Theorem 3, the correction is at most $C\sqrt{qL} + 1$, with $C = 8400$ or 200 and exactly the onsets displayed above. In particular, using $L \le 4\sqrt{q}$ for $q \ge 1$,

$$\frac{\lceil q + C\sqrt{qL} \rceil - q}{q} \le 2Cq^{-1/4} + q^{-1}, \qquad \frac{\sqrt{qL}}{q^{4/5}} \le 2q^{-1/20}. \tag{1}$$

The second ratio is at most $\epsilon > 0$ for $q \ge \max\{1, (2/\epsilon)^{20}\}$; the first is at most ϵ for $q \ge \max\{1, (4C/\epsilon)^4, 2/\epsilon\}$. These quantify the asymptotic language in the title.

Remark (Finite-size scope). The elementary L^2 estimate already gives the tuple failure bound for $k \ge 2q + L + 3$. The theorems above do not improve this sample requirement throughout all their stated ranges. Here are exact sufficient comparison ranges, with all comparisons made at the *minimal* allowed integer k. In Theorem 1,

$$\frac{1000q}{L} \le q \quad \Longleftrightarrow \quad L \ge 1000 \quad \Longleftrightarrow \quad q \ge 2^{1000}.$$

Thus for $q \ge 2^{1000}$ (equivalently $N \ge 2^{2^{1000}}$) its size is at most $\lceil 2q \rceil < 2q + L + 3$. In Theorem 2 the corresponding condition is

$$q \ge 3400^5 = 454\,354\,240\,000\,000\,000,$$

and for Theorem 3(a) it is $q/L \ge 8400^2 = 70\,560\,000$. Below each of these respective conditions the unrounded size, and hence its ceiling, exceeds $2q$; any improvement over $\lceil 2q + L + 3 \rceil$ is then less than $L + 4$, if there is an improvement at all. These are not claimed to be the exact crossover points after the logarithmic term and ceilings are included. For a correction $c(q)$ the exact integer comparison is

$$\lceil q + c(q) \rceil < \lceil 2q + L + 3 \rceil \quad \Longleftrightarrow \quad q + c(q) \le \lceil 2q + L + 3 \rceil - 1.$$

Theorem 3(b) improves the elementary size throughout its range: with $Q_B = 486\,018\,750$, monotonicity of $q/\log_2 q$ gives $q/L \ge Q_B/\log_2 Q_B > Q_B/29 > 40000$, so $200\sqrt{qL} < q$. At the onset of Theorem 1, its size is 53 477 376, whereas the elementary size is 2 097 175. Constants have not been optimized. The transfer to model (U) is proved separately in Section 6.

2. Related work

Erdős and Rényi [2, Theorem 1, pp. 131–132] give simultaneous relative error at most ϵ with probability greater than $1 - \delta$ for independent uniform samples if

$$k \ge 2\log_2 N + 2\log_2(1/\epsilon) + \log_2(1/\delta).$$

With $\epsilon = 1/2$ and $\delta = 1/(2q)$ this is precisely the $2q + L + 3$ benchmark. Erdős and Hall [3, unnumbered Theorem, p. 174] prove simultaneous $(1 \pm \eta)2^k/N$ counts for almost all independent uniform tuples in every finite abelian group, for fixed $\eta > 0$, when

$$k > q\{1 + O_\eta(\log\log q/\log q)\}.$$

This cited historical statement has an unspecified implied constant and no numerical finite onset or failure function. It is not used as an effective premise here. As we read their proof, their Lemma 2, p. 176, drops a product-principal constraint on characters and overcounts binary relations; their Lemma 3 passes to a maximum-count estimate. Our proof instead separates tuples before taking an unconditional second moment.

Hall's 1977 paper [4] is bibliographically confirmed, but its theorem statements were unavailable in the literature search underlying this comparison. We assign no mathematical conclusion to it. This access gap, and incomplete forward-citation coverage, limit any claim about the record.

Stronger results for restricted groups must also be distinguished. For $G = (\mathbb{Z}/2)^d$, the representation function is constant on the span of the sampled elements and zero off it. Erdős and Hall [5, p. 449, Theorem 3 and preceding paragraph] give failure of coverage less than 2^{d-k}; coverage is exact uniformity in this family. Thus $k \geq d + \lceil \log_2(2d) \rceil$ suffices for the tuple failure target here.

Dhar and Dvir [6, Theorem 2.1] establish simultaneous two-sided relative guarantees for finite-field linear hashing. In the prime-field case, for $S \subseteq \mathbb{F}_p^n$, $n \geq 5$, $\tau > 0$, $0 < \delta < 1$,

$$p \geq 32 \max\{n(1+\tau)/(\tau\delta)^2, n\}, \qquad p^r < |S| \leq p^{r+1}, \qquad 4 \leq r \leq n-1,$$

at least a $1-\delta$ fraction of surjective linear maps to $\mathbb{F}_p^{r-3}$ give relative error at most τ at every point. Their Theorems 2.2 and 2.4 give binary variants with logarithmic entropy loss under explicit size hypotheses. These are genuine pointwise balance results, with a finite-field domain.

Lund's preprint [7, Corollary 1.4] gives a further two-sided hashing statement for nonempty $S \subseteq \mathbb{F}_p^n$, $1 \leq d \leq n$, $0 < \tau \leq 1$, and $0 < \delta < 1$. If an integer $a \geq 1$ satisfies

$$p^{a-1} \geq 16(d+a), \qquad \frac{|S|}{p^d} \geq \frac{192 p^a}{\tau^2 \delta},$$

the relative error of every fiber is at most τ with probability greater than $1-\delta$, for either uniform surjections or unrestricted uniform matrices. Taking $S = \{0,1\}^k$, $N = p^d$, $\tau = 1/2$, and $\delta = 1/(2q)$ makes the second condition $2^{k-q} \geq 1536 q p^a$. For each fixed prime this gives logarithmic overhead. We cite the preprint's theorem statement without claiming an independent review of its proof. Neither the field-dependent condition nor the domain supplies a bound uniform over all finite abelian groups, and no without-replacement conclusion is inferred without a transfer argument.

Our quantifier is over *every* finite abelian group, with explicit constants, onsets, and probabilities. Within a literature search of 12 October 2026 we did not locate an explicit bound of this form for all finite abelian groups. This is a bounded search statement, with Hall 1977 unread, and makes no priority claim. The paper does not claim a new solution of the original leading-coefficient question.

3. Proof of Theorem 1

We supply the algebraic ingredients and the Fourier argument in full. Write $H = \widehat{G}$ additively, with identity 0, while evaluating characters multiplicatively: $(\chi+\psi)(x) = \chi(x)\psi(x)$. The structure theorem gives $|H| = N$. For $\Lambda \leq \mathbb{Z}^s$, define

$$d_p(\Lambda) = \dim_{\mathbb{F}_p} \operatorname{im}(\Lambda \longrightarrow \mathbb{F}_p^s), \quad r_p(\Lambda) = s - d_p(\Lambda), \quad r_G(\Lambda) = \max_{p \mid N} r_p(\Lambda).$$

Here $r_p(\Lambda) = \dim_{\mathbb{F}_p}((\mathbb{Z}^s/\Lambda)/p(\mathbb{Z}^s/\Lambda))$. We use "lattice" for any subgroup of $\mathbb{Z}^s$, not necessarily full rank or saturated.

Lemma 1. *For every $\Lambda \leq \mathbb{Z}^s$,*

$$|\operatorname{Hom}(\mathbb{Z}^s/\Lambda, H)| \leq N^{r_G(\Lambda)}. \tag{2}$$

If finitely generated abelian groups satisfy $S \leq K$ and $S \twoheadrightarrow Q$, then for every prime p,

$$\dim(Q/pQ) \leq \dim(S/pS) \leq \dim(K/pK). \tag{3}$$

Proof. Write K as a free abelian group of rank m plus finite torsion. If its p-primary part has b_p cyclic factors, then $\dim(K/pK) = m + b_p$. Each free generator and each p-primary cyclic generator has at most $|H_p|$ images in the p-primary part H_p of H. Torsion at other primes maps to zero, so $|\operatorname{Hom}(K, H_p)| \le |H_p|^{m+b_p}$. Taking $K = \mathbb{Z}^s/\Lambda$ and multiplying over $p \mid N$ proves (2). A subgroup S has free rank at most m, and $S[p]$ embeds in $K[p]$, so its number of p-primary cyclic factors is at most b_p. This proves the second inequality of (3). The first follows from the induced surjection $S/pS \twoheadrightarrow Q/pQ$. □

Lemma 2. *Let $s, B \ge 1$ be integers. A lattice generated by a prescribed set of vectors in $[-B, B]^s \cap \mathbb{Z}^s$ is generated by at most*

$$J(s, B) = s + \lfloor \log_2(s! B^s) \rfloor$$

vectors from that set. Hence there are at most $(2B+1)^{sJ(s,B)}$ such lattices. For lattices generated by binary vectors the number is at most

$$2^{s^2(1+\lceil \log_2 s \rceil)}. \tag{4}$$

Proof. Let d be the rational rank and choose d independent prescribed generators, spanning Λ_0. Projection to d coordinates with nonsingular minor M is injective on the rational span. It follows that

$$[\Lambda : \Lambda_0] \le |\det M| \le d! B^d \le s! B^s.$$

The determinant estimate is the Leibniz expansion, with $d!$ terms of size at most B^d. If the chosen vectors do not yet generate Λ, adjoining one prescribed generator reduces the remaining integer index by a factor at least two. At most $\lfloor \log_2(s! B^s) \rfloor$ additions suffice. Rank zero is immediate. Pad lists with zero vectors for counting; this gives the first count. In the binary case $J(s, 1) \le s(1 + \lceil \log_2 s \rceil)$, and each entry has 2^s choices, which proves (4). □

Lemma 3. *For every lattice $\Lambda \le \mathbb{Z}^s$, writing $r = r_G(\Lambda)$, one has $|\Lambda \cap \{0,1\}^s| \le 2^{s-r}$.*

Proof. Choose $p \mid N$ attaining the maximum r. The image of Λ modulo p has dimension $s - r$. Some $s - r$ coordinate positions give an injective projection on that image, by a nonsingular coordinate minor of a basis. Integer binary vectors stay distinct modulo any prime, including two. Their projections have at most 2^{s-r} binary choices. □

Lemma 4. *If $t \ge 1$ and $\Lambda \le \mathbb{Z}^{2t}$ is generated by binary vectors, then $\Lambda \cap (\mathbb{Z}^t \times \{0\})$ is generated by vectors whose first t coordinates have absolute value at most*

$$B_t = 2^{4t} t!. \tag{5}$$

Proof. List distinct generators as columns of $(X; Y)$, with t rows in each block and $n \le 2^{2t}$ columns. The intersection consists of $(Xz, 0)$ for $z \in \mathbb{Z}^n$ with $Yz = 0$. If $n = 0$ it is zero. Let $r = \operatorname{rank}_{\mathbb{Q}} Y$. If $r = 0$, coordinate vectors generate the kernel and their images have coordinates at most one. If $n = r$, the kernel is zero. Otherwise choose r columns and r rows giving a nonsingular matrix C, and write F for the remaining columns on those rows. The selected rows are independent, since C is invertible, and span the row space of Y, since its rank is r. With the pivot coordinates first, the full kernel condition is therefore

$$z = (x, y), \qquad x = -C^{-1} F y.$$

Put $D_0 = |\det C|$, so $1 \le D_0 \le r!$. The allowed integer y form a lattice T containing $D_0 \mathbb{Z}^{n-r}$. Reducing coordinates modulo D_0 shows that T is generated by $D_0 e_j$ and its elements in $\{0, \dots, D_0 - 1\}^{n-r}$. Every adjugate entry of C has absolute value at most $(r-1)!$. For a representative in this box, each coordinate of x is at most

$$\frac{r(r-1)!(n-r)(D_0 - 1)}{D_0} \le n r!.$$

For $y = D_0 e_j$ the bound is $r!$. The y coordinates are also at most $D_0 \le r!$. Thus the integer kernel is generated by z with $\|z\|_\infty \le nr!$. Every coordinate of Xz is at most $n^2 r! \le 2^{4t} t!$, as required. □

For $u = (u_1, \dots, u_t) \in H^t$ and $\theta \in H^{2t}$ set

$$\Lambda_B(u) = \left\langle c \in [-B_t, B_t]^t \cap \mathbb{Z}^t : \sum_j c_j u_j = 0 \right\rangle,$$

$$\Lambda_C(\theta) = \left\langle e \in \{0,1\}^{2t} : \sum_j e_j \theta_j = 0 \right\rangle.$$

Lemma 5. *If the first t entries of θ are u, then $r_G(\Lambda_B(u)) \le r_G(\Lambda_C(\theta))$.*

Proof. Identify the intersection of $\Lambda_C(\theta)$ with the first coordinate block with a lattice $I \le \mathbb{Z}^t$. Every vector in $\Lambda_C(\theta)$ is an actual relation. Lemma 4 therefore generates I by bounded relations of u, and $I \le \Lambda_B(u)$. The coordinate inclusion gives an injection $\mathbb{Z}^t/I \hookrightarrow \mathbb{Z}^{2t}/\Lambda_C(\theta)$, and there is a surjection $\mathbb{Z}^t/I \twoheadrightarrow \mathbb{Z}^t/\Lambda_B(u)$. Apply (3) for each $p \mid N$ and take maxima. The maximizing primes need not coincide. □

3.1. **Fourier coefficients and a common event.** In model (I), let

$$a_\chi = \prod_{i=1}^{k} \frac{1 + \chi(g_i)}{2}, \qquad h = k - q > 0.$$

Orthogonality gives

$$D(g) = \sum_{\chi \ne 0} a_\chi \overline{\chi(g)}. \tag{6}$$

Indeed expanding the product and then summing all characters detects each subset sum. A nonprincipal character has sum zero over G, because translation by an element on which it is not one both preserves and nontrivially multiplies that sum. All sums here are finite.

Let $\rho = 2^{-q/16}$ and define

$$E = \{a_\chi = 0 \text{ for every } \operatorname{ord}(\chi) = 2\} \cap \{|a_\chi| \le \rho \text{ for every } \operatorname{ord}(\chi) \ge 3\}.$$

An order-two coefficient is zero or one and is one with probability 2^{-k}. Its union bound is at most $N 2^{-k} = 2^{-h}$. If $\operatorname{ord}(\chi) \ge 3$, then for uniform $x \in G$

$$\mathbb{E}\left|\frac{1+\chi(x)}{2}\right|^4 = \frac{1}{16}\mathbb{E}(6 + 4\chi(x) + 4\chi(x)^{-1} + \chi(x)^2 + \chi(x)^{-2}) = \frac{3}{8}.$$

All four nonconstant characters in the expansion are nonprincipal. Independence and Markov's inequality now give

$$\mathbb{P}_{\mathrm{I}}(\exists \chi,\ \operatorname{ord}(\chi) \ge 3 :\ |a_\chi| > \rho) \le N \rho^{-4} (3/8)^k \le 2^{5q/4} (3/8)^q \le 2^{-q/8}.$$

The last step uses $\log_2(8/3) > 11/8$, equivalently $3^8 = 6561 < 8192 = 2^{13}$. Consequently

$$\mathbb{P}_{\mathrm{I}}(E^c) \le 2^{-h} + 2^{-q/8}. \tag{7}$$

3.2. **A split of the Fourier power.** Fix integers $t \ge 6$, $1 \le a \le t$, and let $\ell = \lceil \log_2 t \rceil$. Expand (6) to obtain $D(g)^t = P_{\mathrm{low}}(g) + P_{\mathrm{high}}(g)$, where P_{low} collects the ordered nonprincipal tuples u with $r_G(\Lambda_B(u)) < a$, and P_{high} collects the rest. Multiplicities are those of ordered tuples. Lemma 2 bounds the number of possible $\Lambda_B(u)$ by 2^{A_t}, where

$$A_t = 2t^4(6 + \ell)^2 \le 8t^6. \tag{8}$$

To see every factor, use

$$\log_2 B_t \le t(4 + \ell),$$

$$J(t,B_t)\le t+t\ell+t^2(4+\ell)\le t^2(5+2\ell),$$
$$t\log_2(2B_t+1)\le t(\log_2 B_t+2)\le t^2(6+\ell).$$

The product is at most $2t^4(6+\ell)^2$. Since $\ell\le t$ and $6+\ell\le 2t$ for $t\ge 6$, (8) follows. For a fixed low pattern Λ, the tuples are among the homomorphisms from $\mathbb{Z}^t/\Lambda$ to H. Writing $r=r_G(\Lambda)<a$, their number is at most $N^r\le N^{a-1}\le N^a$. Thus on E, for every g,

$$|P_{\mathrm{low}}(g)|\le 2^{A_t+aq-tq/16}. \tag{9}$$

The crucial second moment is unconditional:

$$\mathbb{E}|P_{\mathrm{high}}(g)|^2\le 2^{C_t-ah},\qquad C_t=4t^2(2+\ell)\le 8t^3. \tag{10}$$

Indeed for two high tuples u,v, set $\theta=(u,-v)$. Independence and orthogonality give the exact nonnegative expectation

$$\mathbb{E}\left[\prod_j a_{u_j}\prod_j \overline{a_{v_j}}\right]=\left[2^{-2t}\#\{e\in\{0,1\}^{2t}:\sum_j e_j\theta_j=0\}\right]^k. \tag{11}$$

The phase from evaluating at g has absolute value one. The triangle inequality therefore bounds the expanded second moment by the sum of the right sides of (11). Group pairs by $\Lambda=\Lambda_C(\theta)$ and put $r=r_G(\Lambda)$. The binary vectors of Λ are exactly the binary relations of θ: all generators are relations and all binary relations were included. Lemma 3 bounds the bracket by 2^{-r}, and Lemma 5 gives $r\ge a$. By (2) at most N^r tuples θ, hence at most N^r pairs (u,v), have this pattern. Its contribution is at most $N^r2^{-rk}=2^{-rh}\le 2^{-ah}$. Lemma 2 gives at most 2^{C_t} binary patterns, since $\lceil\log_2(2t)\rceil=1+\lceil\log_2 t\rceil$. Finally $2+\ell\le 2t$ for $t\ge 2$, proving (10). The event E is absent from this expectation. No signed polynomial is treated as a nonnegative random variable.

3.3. **The parameters for Theorem 1.** Assume first $L\ge 500$ and take $t=\lfloor L\rfloor$, $a=\lfloor t/32\rfloor$. For every integer $t\ge 64$,

$$512t^5\le 2^t. \tag{12}$$

At $t=64$ the left side is $2^{39}\le 2^{64}$. Its successive ratio is at most $(65/64)^5<(9/8)^5<2$, the last comparison being $59049<65536$. Induction proves (12). Since $q=2^L\ge 2^t$, it follows that $A_t\le 8t^6\le tq/64$, and

$$|P_{\mathrm{low}}(g)|\le 2^{-tq/64}\le 2^{-t-1}\quad(g\in G)\quad\text{on } E.$$

Here $a\le t/32$, $q\ge 128$, and $2t\ge t+1$ justify the exponents. Also

$$C_t+2t+2\le 8t^3+2t+2\le 16t^3\le 512t^5\le q.$$

As $t\ge 500$, we have $a\ge t/32-1\ge t/40$, and

$$ah\ge\frac{t}{40}\frac{1000q}{L}\ge\frac{25tq}{t+1}\ge 24q,$$

where $L<t+1$ and $t\ge 24$. If E holds and $|D(g)|>1/2$, then $|P_{\mathrm{high}}(g)|>2^{-t-1}$. Markov's inequality for $|P_{\mathrm{high}}(g)|^2$ and a union bound over N points therefore give

$$\mathbb{P}_{\mathrm{I}}(\mathcal{F})\le 2^{-h}+2^{-q/8}+2^{q+2t+2+C_t-ah}\le 2^{-h}+2^{-q/8}+2^{-22q}. \tag{13}$$

For real $x\ge 4$, $2^x\ge x^2$: equality holds at four and the logarithmic derivative is $\ln 2-2/x>0$ thereafter, since $\ln 2>1/2$. Thus $q\ge L^2$. For $L\ge 20$, $L^2\ge 8(L+3)$, by its value at twenty and positive derivative of the difference. We obtain $h\ge 1000q/L\ge 1000L\ge L+3$ and $q/8\ge L+3$. Each term in (13) is consequently at most $1/(8q)$, so the sum is at most $3/(8q)$.

For $20 \le L \le 500$ the stated size gives $k \ge 3q$. Orthogonality gives $\mathbb{E}|a_\chi|^2 = 2^{-k}$ for every non-principal character. Parseval in (6), followed by Markov's inequality, gives the general elementary estimate

$$\mathbb{E}\sum_g |D(g)|^2 = N(N-1)2^{-k}, \qquad \mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le 4\,2^{2q-k}. \tag{14}$$

In this range it is at most $4\,2^{-q} \le 1/(2q)$, since $q \ge 2^{20}$ and $2^q \ge q^2 \ge 8q$. The two ranges cover all $L \ge 20$, including their common boundary. This proves the tuple part of Theorem 1; Section 6 proves the subset part. Notice also the general consequence of (14):

$$k \ge 2q + L + 3 \quad \Longrightarrow \quad \mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le \frac{1}{2q}. \tag{15}$$

4. A sharper lattice count and Proposition M

Lemma 2′. *For integers $s, B \ge 1$, the number of lattices in $\mathbb{Z}^s$ generated by vectors of $[-B, B]^s \cap \mathbb{Z}^s$ is at most*

$$(s+1)2^s(2B+1)^{s^2}(s!B^s)^s. \tag{16}$$

Proof. Let $d > 0$ be the rank. Choose independent box vectors $v_1, \dots, v_d$ in Λ and a set S of d coordinates giving a nonsingular minor M. Put $V = \mathbb{Q}\Lambda$ and $\Lambda_0 = \sum_i \mathbb{Z}v_i$. Coordinate projection π restricts to an isomorphism $V \to \mathbb{Q}^d$: it takes the chosen basis to independent vectors. The finite group

$$T_0 = \mathbb{Z}^d/\pi(\Lambda_0)$$

has order $D_0 = |\det M| \le d!B^d$ and is generated by d elements. Each subgroup U of T_0 is also generated by at most d elements. Indeed (3) bounds its number of p-primary cyclic factors by d for every prime. Pad the primary decompositions to d factors and combine across primes by the Chinese remainder theorem. Thus U has a generating ordered d-tuple in T_0^d, and there are at most D_0^d possible U.

The data $(d, v_1, \dots, v_d, S, U)$, with $U = \pi(\Lambda)/\pi(\Lambda_0)$, determine Λ. They determine V and the full inverse image $\pi(\Lambda)$ of U, and

$$\Lambda = \{x \in V : \ \pi(x) \in \pi(\Lambda)\}.$$

For if $\pi(x) = \pi(y)$ with $y \in \Lambda$, injectivity on V gives $x = y$; integrality follows as well. Unrealized data can only overcount. For fixed d there are at most $(2B+1)^{sd}2^s(d!B^d)^d$ data, which is at most the same expression with $d = s$. Sum over $d = 0, \dots, s$, including the unique zero lattice, to prove (16). □

For $t \ge 3$, $\ell = \lceil \log_2 t \rceil$, and B_t from (5), this gives at most $2^{A'_t}$ patterns $\Lambda_B(u)$, where

$$A'_t = t^3(10 + 2\ell). \tag{17}$$

In fact $t! \le 2^{t\ell}$, $\log_2 B_t \le 4t + t\ell$, and $2B_t + 1 \le 4B_t$, so the logarithm of (16) is at most

$$t^3(8 + 2\ell) + t^2(2 + \ell) + t + \log_2(t+1).$$

For integer $t \ge 3$, $\ell \le t$ and $\log_2(t+1) \le t$; the latter follows by induction from $t + 1 \le 2^t$. The remainder is at most $t^3 + 2t^2 + 2t \le 2t^3$, because $t^2 \ge 2t + 2$ for $t \ge 3$. This proves (17).

Proposition M. *Suppose $q \ge 128$, k is an integer, $h = k - q > 0$, and integers $t \ge 6$, $1 \le a \le t$, with $\ell = \lceil \log_2 t \rceil$, satisfy*

$$A'_t + aq + t + 1 \le tq/16, \tag{P1}$$

$$q + 2t + 2 + C_t + 3 + L \le ah, \tag{P2}$$

$$h \ge 3 + L. \tag{P3}$$

Then

$$\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le 2^{-h} + 2^{-q/8} + \frac{1}{8q} \le \frac{3}{8q} < \frac{1}{2q}.$$

Proof. Use exactly the event and split of Section 3. Replace the low-pattern count by (17). On E, (P1) gives $|P_{\mathrm{low}}(g)| \le 2^{A'_t + aq - tq/16} \le 2^{-t-1}$. The unconditional estimate (10) is unchanged, since it uses the binary-pattern count. Markov's inequality and the union bound therefore give

$$\mathbb{P}_{\mathrm{I}}(E \cap \mathcal{F}) \le 2^{q+2t+2+C_t-ah} \le 2^{-3-L} = \frac{1}{8q}$$

by (P2). Combine with (7). Condition (P3) bounds the first exponential by $1/(8q)$. For real $q \ge 128$, $q/8 \ge 3 + \log_2 q$: it holds at 128 since $16 > 10$, and the derivative of the difference is $1/8 - 1/(q \ln 2) > 0$. This bounds the second exponential too. □

The conditions involving h are monotone in h. This will give the "every integer k" assertion of Theorem 3.

5. Parameters and exact arithmetic for Theorems 2 and 3

We use the following elementary facts for real variables. For $x \ge 1$, $\log_2 x \le 4\sqrt{x}$, since $\ln x = 2 \ln \sqrt{x} \le 2(\sqrt{x} - 1) \le 2\sqrt{x}$ and $1/\ln 2 < 2$. The function $2^x/x^2$ increases for $x \ge 64$, as its logarithmic derivative is $\ln 2 - 2/x > 0$. The functions $x/\log_2 x$ and $x^{3/5}/\log_2 x$ increase for $x \ge e$ and $x \ge e^{5/3}$, respectively: the signs of their derivatives are those of $\ln x - 1$ and $(3/5)\ln x - 1$. Also $q \ge 3 + \log_2 q$ for $q \ge 8$, by checking eight and the positive derivative of $q - \log_2 q$. Finally, for integer $t \ge 2$,

$$2^{\ell(t)-1} < t \le 2^{\ell(t)}, \qquad \ell(t+1) \le \ell(t) + 1, \qquad \ell(t) = \lceil \log_2 t \rceil.$$

The step bound follows from $t + 1 \le 2t$.

For transparency all numerical endpoint comparisons used below are listed in Table 1. Every entry is an integer inequality. It can be checked by multiplication alone; the supplied program also checks it. A positive margin in a row means left side minus right side for the displayed $\ge$ comparison. Define

$$Q_1 = 512 \cdot 160^5 = 53\,687\,091\,200\,000, \qquad Q_B = 49 \cdot 575^2 \cdot 30 = 486\,018\,750.$$

Label	Exact comparison
A1	$3400^5\,160^5 \ge 512 \cdot 960^5\,161^5$
A2	$3400^5 \ge 32 Q_1$
A3	$2^{64} \ge 3402^2\,64^2$
B0	$2^{28} \le Q_B < 2^{29}$, $\ell(575) = 10$
B1	$25(575 - 23) = 24 \cdot 575$
B2	$200^2\,49^2\,575^2 \ge 1300^2 \cdot 56 \cdot 576^2$
B3	$8400^2 \cdot 28 \ge 4 Q_B$
B4	$2^{64} \ge 8402^2\,64^2$
B5	$Q_B \ge 343^2$
B6	$49 \cdot 4^9 \cdot 30 > 2^{28}$
C1	$9^5 Q_B^3 \ge (40000 \cdot 29)^5$

Table 1. Exact endpoints. The margins in B2 and B3 are 353 944 360 000 and 31 605 000; the margin in C1 is 4 678 751 774 701 839 942 138 671 875 000.

5.1. **Theorem 2.** Let $q \ge 64$ and $h = k - q \ge 3400q^{4/5}$. If $q < Q_1$, A2 gives $3400^5 \ge 32Q_1 > 32q$, hence $h \ge 2q$. It follows that $k - 2q \ge q \ge 3 + L$, so (15) applies.

If $q \ge Q_1$, take the largest integer t with $512t^5 \le q$. Then $t \ge 160$ and $q < 512(t+1)^5$. Put $a = \lfloor t/32 \rfloor$; then $t/40 \le a \le t/32$, since $\lfloor t/32 \rfloor > t/32 - 1 \ge t/40$ for $t \ge 160$. The correction guarantees

$$h \ge 960q/t. \tag{18}$$

Indeed maximality, $(t+1)/t \le 161/160$, and A1 give

$$960^5 q < 512 \cdot 960^5 (t+1)^5 \le 512 \cdot 960^5 (161/160)^5 t^5 \le 3400^5 t^5.$$

Taking positive fifth roots and rearranging gives $960q/t < 3400q^{4/5} \le h$.

Now $10 + 2\ell \le 8t^3$ (use $\ell \le t$ and $t \ge 3$), so

$$A'_t \le 8t^6 \le tq/64, \qquad aq \le tq/32, \qquad t + 1 \le 2t \le tq/64.$$

Their sum proves (P1). Next

$$C_t + 2t + 2 \le 16t^3 \le 512t^5 \le q, \qquad 3 + L \le q.$$

Thus the left side of (P2) is at most $3q$, while (18) gives $ah \ge 24q$. Finally $t^2 \le t^5 \le 512t^5 \le q$ implies $t \le \sqrt{q}$, and $h \ge 960\sqrt{q} \ge 4\sqrt{q} + 3 \ge L + 3$. This proves (P3). All hypotheses of Proposition M hold, proving the tuple assertion. The subset assertion follows in Section 6.

5.2. **Theorem 3(b): the main branch.** For positive integers t define

$$f(t) = 49t^2(10 + 2\ell(t)).$$

It is strictly increasing and unbounded. Suppose $q \ge Q_B = f(575)$, and choose maximal t with $f(t) \le q$. Then $t \ge 575$, $\ell \ge 10$, and $q < f(t+1)$. Set $a = \lfloor t/24 \rfloor$. Since an integer has remainder at most 23 modulo 24,

$$t/25 \le (t - 23)/24 \le a \le t/24; \tag{19}$$

the first inequality is exactly $t \ge 575$.

For (P1), use

$$tq/48 \ge (49/48)t^3(10 + 2\ell) = A'_t + A'_t/48 \ge A'_t + t + 1.$$

The last step holds for $t \ge 3$: $A'_t \ge 14t^3$, $t + 1 \le 4t/3$, and $14t^3/(t+1) \ge (21/2)t^2 \ge 189/2 > 48$. Adding $aq \le tq/24$ proves (P1).

For (P2),

$$C_t = 4t^2(2 + \ell) \le 2t^2(10 + 2\ell) \le 2q/49.$$

Directly from $q \ge f(t) \ge t^2$ we have $t \le \sqrt{q}$; no stronger hypothesis on q is invoked. By B5, $\sqrt{q} \ge 343$, so

$$2t + 5 + L \le 6\sqrt{q} + 5 \le 7\sqrt{q} \le q/49.$$

The left side of (P2) is at most $52q/49$. In view of (19), it suffices that

$$h \ge \frac{1300}{49}\frac{q}{t}. \tag{20}$$

This also gives (P3), since $h \ge 26\sqrt{q} \ge 4\sqrt{q} + 3 \ge L + 3$.

We deduce (20) from $h \ge 200\sqrt{qL}$. Since $t > 2^{\ell-1}$ and $\ell \ge 10$,

$$q \ge f(t) > 49 \cdot 4^{\ell-1}(10 + 2\ell) \ge \frac{735}{2}\, 4^\ell > 2^{2\ell+8}.$$

Thus $L > 2\ell + 8 \ge 28$. Maximality of t now gives

$$\begin{aligned} q &< 49(t+1)^2(10 + 2\ell(t+1)) \\ &\le 49(t+1)^2(12 + 2\ell) < 56(t+1)^2 L \le 56(576/575)^2 t^2 L. \end{aligned}$$

Here $12+2\ell < L+4 < (8/7)L$, using $L > 28$. By B2,

$$(1300/49)^2 q < 40000 t^2 L.$$

Multiply by q/t^2 and take positive square roots to obtain $(1300/49)q/t < 200\sqrt{qL} \le h$. Proposition M proves $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \le 3/(8q)$ for every integer $k \ge q + 200\sqrt{qL}$.

5.3. **Theorem 3(a): the remaining onset range.** If $64 \le q < Q_B$, monotonicity and B0 give

$$q/L < Q_B/\log_2 Q_B \le Q_B/28.$$

By B3, $8400^2 L \ge 4q$, so $h \ge 8400\sqrt{qL}$ implies $h \ge 2q$. Therefore $k - 2q \ge q \ge 3 + L$, and (15) applies. For $q \ge Q_B$ use the main branch and $8400 \ge 200$. This proves the tuple assertion for every required integer k. The endpoint $q = Q_B$ belongs to the main branch, so the ranges leave no gap.

5.4. **Corollary C.** For $k = \lceil q + 3q^{4/5} \rceil$ it suffices to prove $3q^{4/5} \ge 200\sqrt{qL}$, equivalently $9q^{3/5} \ge 40000L$. The ratio $q^{3/5}/\log_2 q$ is increasing for $q \ge Q_B$. At $q = Q_B$, C1 and $\log_2 Q_B < 29$ imply

$$9Q_B^{3/5} \ge 40000 \cdot 29 > 40000 \log_2 Q_B.$$

Thus the sufficient inequality holds throughout this range. The ceiling preserves the lower bound on k, so Theorem 3(b) gives the tuple conclusion. The required upper bound for the subset conclusion is verified next.

6. Transfer from independent sampling to a uniform subset

For independent uniform samples the collision probability is at most

$$\binom{k}{2}/N = \frac{k(k-1)}{2N}.$$

If $k(k-1) \le N$, distinctness therefore has probability at least $1/2$. For nonnegative integer k and $N \ge 2$ this condition also implies $k \le N$: $k \ge N+1$ would give $k(k-1) \ge N(N+1) > N$. Conditional on distinctness, the ordered sample is uniform on all ordered distinct tuples. Forgetting order gives exactly model (U), and the indexed subset counts become the subset counts of that set. Balance is invariant under ordering, so

$$\mathbb{P}_{\mathrm{U}}(\mathcal{F}) = \mathbb{P}_{\mathrm{I}}(\mathcal{F} \mid \text{distinct}) \le \frac{\mathbb{P}_{\mathrm{I}}(\mathcal{F})}{\mathbb{P}_{\mathrm{I}}(\text{distinct})} \le 2\mathbb{P}_{\mathrm{I}}(\mathcal{F}). \tag{21}$$

In particular $k \le 2^{q/2}$ is a sufficient upper bound.

For Theorem 1, $L \ge 20$ and

$$k \le q + 1000q/L + 1 \le 51q + 1 \le 52q.$$

For real $x \ge 16$, $2^x \ge x^4$: equality holds at sixteen and the logarithmic derivative $\ln 2 - 4/x$ is positive thereafter. Since $q \ge 2^{20} \ge 52$, it follows that $2^q \ge q^4 \ge 2704q^2 \ge k^2$. This proves both $k \le N$ and the distinctness condition, completing Theorem 1.

For Theorem 2, $k \le 3402q$ because $q^{4/5} \le q$ and $1 \le q$. By A3 and monotonicity of $2^q/q^2$ for real $q \ge 64$, $(3402q)^2 \le 2^q$, so (21) applies. For Theorem 3, the stipulated upper bound on k is sufficient. Its interval is nonempty: $L \le q$ and B4 imply

$$\lceil q + 8400\sqrt{qL} \rceil \le 8402q \le 2^{q/2} \qquad (q \ge 64).$$

The assertion for constant 200 follows a fortiori. Finally, in Corollary C, $k \le 5q \le 2^{q/2}$, using the same bound with 8402. This proves all the subset conclusions. No conclusion for arbitrary larger k in model (U) is needed or asserted.

7. Scope and limits

Remark 1 gives the finite-size comparison with L^2. The group-uniform statements include elementary abelian groups, cyclic groups of every order, and mixed-primary groups: r_G takes all primes dividing N into account. In particular, neither a rational-rank approximation nor a prime-exponent hypothesis is implicit in the proof.

Proposition (A limit of the displayed sufficient conditions)**.** *If $q \geq 128$, $t \geq 6$, $a \geq 1$, and $h > 0$ satisfy* (P1)–(P2)*, then*

$$h > 16q/t + 64t(2+\ell) \geq 64\sqrt{q(2+\ell)} \geq 128\sqrt{q}. \tag{22}$$

Proof. The positive terms in (P1) give $a < t/16$. Equation (P2) gives $ah > q + C_t$. Since $a > 0$, $h > (q + C_t)/a > 16(q + C_t)/t = 16q/t + 64t(2 + \ell)$. Apply $x + y \geq 2\sqrt{xy}$, and use $\ell \geq 2$. □

This is only a limitation of these sufficient conditions. It is not a lower bound for sharper estimates using realized patterns, their ranks, their weights, or extra structure of the group.

There is also a much smaller necessary correction for a theorem uniform over groups in model (I). For $G = (\mathbb{Z}/2)^n$, $N \geq 4$, and $h = k - q \geq 1$, let E_χ be the event that a nonzero character annihilates every sample. It forces failure, since all sums then lie in a proper kernel. Each E_χ has probability 2^{-k}, and two distinct nonzero characters have joint kernel of index four, so $\mathbb{P}_{\mathrm{I}}(E_\chi \cap E_\psi) = 2^{-2k}$. Bonferroni gives

$$\begin{aligned}\mathbb{P}_{\mathrm{I}}(\mathcal{F}) &\geq (N-1)2^{-k} - \binom{N-1}{2}2^{-2k} \\ &\geq (N-1)2^{-k}(1 - 2^{-h}/2) \geq (3/8)2^{-h}.\end{aligned}$$

The last bound uses $(N-1)/N \geq 1/2$ and $1 - 2^{-h}/2 \geq 3/4$. Consequently $\mathbb{P}_{\mathrm{I}}(\mathcal{F}) \leq 1/q$ forces

$$h \geq \log_2 q - \log_2(8/3).$$

This argument concerns model (I) on elementary abelian two-groups. It is neither a fluctuation claim for every group nor a lower bound proved here for model (U).

7.1. **Heuristic discussion (not used in the theorems).** One motivation for a possible square-root scale is the balance between the spatial factor 2^q and the number of relation patterns. A typical-coefficient heuristic would suggest $|a_\chi|$ cannot all be much below $2^{-k/2}$ and hence that a low-rank cutoff must have a comparable to at most $t/2$. This heuristic is false for balanced samples in elementary abelian two-groups, where every nonprincipal coefficient vanishes. Even in other groups it is not proved here. Combining the heuristic cutoff with the spatial cost suggests h of order q/t. There are at least 2^{t^2} binary-generated lattices in $\mathbb{Z}^{2t}$: the 2^{t^2} graph subspaces $\{(x, Mx) : x \in \mathbb{F}_2^t\}$, as M varies, have distinct mod-two images, and their binary lifts generate distinct lattices. Paying for all such patterns suggests a second cost of order t in h. Balancing these two suggested costs gives a square-root scale, while the bounds actually used here include a logarithmic loss. The count of all lattices does not establish a lower bound on the number or weight of the high patterns realized for a fixed group. Thus this paragraph proves no barrier beyond (22).

The large constant 8400 also reflects the chosen hand-over to the L^2 branch. This is a comment about that strategy, not a lower bound on constants in other parameter choices. For example $f(48) = 2\,483\,712$ and the same cutoff would have $a = \lfloor 48/24 \rfloor = 2$, whereas just below $f(48)$ maximality gives $t \leq 47$ and $a \leq 1$. Such small cutoffs make this particular hand-over costly. No optimized universal threshold is claimed.

7.2. **Obstructions encountered in simpler approaches.** These observations explain the ingredients used above; none is an impossibility theorem for alternative proofs.

(1) Direct central moments with a union bound retain the terms $(N-1)2^{-k}$ from a repeated annihilating character in elementary two-groups. The additional spatial factor N prevents that estimate from closing for $h \ll q$. Likewise (14) alone gives $4\,2^{q-h}$, which is uninformative in that regime.

(2) Fresh averaging by a random translate gives exact L^2 contraction. For nonnegative mean-one f, put $S_p(f) = N^{-1}\sum_g f(g)^p$ and let $T_x f$ denote translation. Binomial expansion gives

$$\mathbb{E}_x S_p((f+T_x f)/2) = 2^{1-p}S_p(f) + 2^{-p}\sum_{j=1}^{p-1}\binom{p}{j}S_j(f)S_{p-j}(f).$$

For $p \geq 3$, Hölder interpolation gives $S_j(f) \leq S_{p-1}(f)^{(j-1)/(p-2)}$. Hence the sum term is at most $(1-2^{1-p})S_{p-1}(f)$. This recurrence still propagates the second-moment rate $1/2$ and supplied no needed maximum estimate.

(3) Taking absolute values before orthogonality loses the exact binary-relation identity (11). The elementary Cauchy–Schwarz substitute only gives a square-root relation probability, losing the required cancellation between N^r and 2^{-rk}.

(4) A signed remainder cannot be bounded by first-moment Markov. In C_5, four characters with exponents $(1,1,1,2)$ have principal product, but at a generator the product of their $1+\chi$ factors is $-16\cos^3(\pi/5)\cos(2\pi/5) < 0$. The squared modulus in (10) avoids that problem.

(5) Binary or signed-binary relations within u need not preserve rank after combining tuples. If u_0 has order five, $(u_0, 2u_0)$ has no nonzero relation in $\{-1,0,1\}^2$. But $(u_0, 2u_0, -u_0, -u_0)$ has binary relations $(1,0,1,0), (1,0,0,1), (0,1,1,1)$; their quotient is infinite cyclic, and their coordinate intersection contains $(2,-1,0,0)$. The respective rank parameters are two and one. The enlarged box in Lemma 4 is essential to this argument.

(6) Counting arbitrary subsets of the binary cube costs $2^{2^{2t}}$. Lemma 2 replaces this by an integer lattice count. Rational rank alone is also inadequate: $\mathbb{Z}/2\mathbb{Z}$ has rational rank zero but has N homomorphisms into $(\mathbb{Z}/2)^n$.

(7) The common coefficient bound alone gives only $|D(g)| \leq (N-1)N^{-1/16}$. In an elementary three-group the expected Fourier ℓ^1 norm is $(N-1)(2/3)^k$, also too large near $k = q$. A split into two sample blocks followed by Cauchy–Schwarz requires small Fourier ℓ^2 error in both blocks and did not retain the leading coefficient one.

(8) In primal subset-incidence moments, torsion induces dependencies beyond repeated subsets, already in elementary two-groups. An independent-row approximation therefore fails; (11) keeps those dependencies exactly. Higher even moments of P_{high} were not developed into a stronger bound: combining $2m$ tuples gives length $2mt$, but the available rank guarantee is still only a, while the binary-pattern exponent grows with that length. No correction below the proved square-root logarithmic scale is established here.

8. Reproduction and finite checks

The mathematical proof is analytic and uses only the finite integer comparisons explicitly displayed above. The companion programs are checks, not a sampling experiment at the enormous onsets and not a formal proof. They use integer group operations and exact modular elimination, JavaScript `BigInt`, or Python integers and `fractions.Fraction`; no solver or floating-point acceptance criterion is used. The two programs are identical to the reviewed versions except for the removal of an internal output path and of project labels in comments; the archived JSON outputs

are unchanged. Their short filenames retain their original check-series labels; these labels play no mathematical role.

The first program, `cr19-exact-checks.mjs`, reports 7072 assertions. For each of

$$C_2, C_3, C_4, C_5, C_6, C_7, C_8, C_9, C_{10}, C_{12}, C_2^2, C_2^3, C_3^2, C_2 \times C_4, C_2 \times C_6,$$

it enumerates all ordered character tuples of lengths two and four, including principal characters for pattern checks. It verifies the cube bound and the count of tuples per pattern. The binary relation set is an exact key for the generated relation lattice, since the lattice's intersection with the binary cube is exactly that set. It checks the fourth-moment relation counts $16, 8, 6$ at character orders one, two, and at least three. It also enumerates every ordered sample of sizes $k = 0, 1, 2, 3$ and checks, for $s = 2, 4$, the identities

$$\sum_A (NR_A(0) - 2^k)^s = N^k \sum_{u \in (H\setminus\{0\})^s} r(u)^k,$$
$$\sum_A \sum_g (NR_A(g) - 2^k)^s = N^{k+1} \sum_{\substack{u \in (H\setminus\{0\})^s \\ \sum_j u_j = 0}} r(u)^k,$$

where $r(u) = \#\{e \in \{0,1\}^s : \sum_j e_j u_j = 0\}$. It checks the three integer bases for the spectral estimate and induction, (12) for $64 \le t \le 2048$, and $\lfloor t/32 \rfloor \ge t/40$ for $500 \le t \le 2048$. These sweeps supplement the general arguments, and do not check every onset inequality of Theorem 1 by themselves.

The second program, `cr19b_checks.py`, reports 53 325 assertions. It checks Table 1, the two onset parameters, the elementary integer inequalities for $3 \le t \le 5000$ (factorials through 1000), and both transfer comparisons for integer $64 \le q \le 4096$. It enumerates all box-generated lattices by closure under adjoining a box vector, with canonical row Hermite normal forms. The exact totals are

(s, B)	$(1,1)$	$(1,2)$	$(1,3)$	$(2,1)$	$(2,2)$	$(3,1)$
lattices	2	3	4	7	31	64

and all satisfy Lemma $2'$. It also evaluates (P1)–(P3) at $q = Q_1 m$ for $m \in \{1, 2, 7, 10^3, 10^6\}$, and at $q = Q_B m$ for $m \in \{1, 2, 3, 10, 10^3, 10^6, 10^{12}\}$, using integer logarithm bounds and integer square roots in the conservative direction. These sample parameters are checks only; the proofs above cover all real q in the stated ranges. The minimal positive integer constants passing the isolated comparisons A1, B2, and B3 are 3364, 199, and 8333. This is not a claim of optimal theorem constants.

From the bundle directory run

```
UV_THREADPOOL_SIZE=1 node --single-threaded code/run-checks.mjs results
```

with Node.js and Python 3 available. The runner executes scratch copies, redirects the first program's legacy output destination, and compares the results with the archived JSON after removing timing fields. It never changes the archived JSON outputs. The checks are sequential and use one computation thread. The stored original runtimes are about 0.12 and 50.34 CPU seconds; times vary by machine. The second program's last sample uses a long linear parameter search, which affects runtime but not the argument. The README gives individual commands and build instructions. `SHA256SUMS` records the publication files.

AI use and verification status

The mathematical development and manuscript preparation used GPT-6 Astra (OpenAI) and Claude Opus 5.5 (Anthropic). The first bound was developed using GPT-6 Astra and received two independent Claude proof reviews with verdict PASS, together with a separate exact-arithmetic verification. The sharper lattice count and subsequent bounds were developed using Claude and

received two independent GPT-6 Astra reviews with verdict PASS-WITH-REPAIRS, and a separate numerical verification with verdict PASS. The repairs replaced an inapplicable parameter citation by a direct inequality, restricted the methodological barrier to the sufficient conditions actually proved, and clarified ceilings and the scope of the lower-bound example. A subsequent delta verification returned PASS; theorem statements, constants, and onsets were unchanged. All remaining editorial points are incorporated here. The exact checks shipped with the paper are the archived author programs and outputs; reviewer programs are not part of this bundle. These reviews were performed using AI systems, not human peer review. The present manuscript was assembled from the reviewed reports and checked again against their arguments. The literature comparison is bounded to the search of 12 October 2026 described in Section 2. No Lean or other proof-assistant formalization of these explicit bounds is supplied.

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